

BotFlow Exchange

A safer swap interface for BOT Chain

WHITEPAPER v1.0 - October 2026

Live on BOT Chain Testnet (chain ID 968)

<https://botflow-woad.vercel.app/>

*BotFlow Exchange is an independent interface and is not affiliated
with BOT Chain or BDEX. Test tokens have no real value.*

1. Abstract

BotFlow Exchange is an open source swap app for BOT Chain. It combines a browser based web app with an on chain router contract, BotFlowRouter, that sits in front of BDEX V3 liquidity. Users can pay with or receive native BOT without separate wrap steps, see the full route and cost before signing, and rely on simulation plus a fresh price check to stop bad trades before they cost gas. The project is currently live on BOT Chain Testnet (chain ID 968) and is built to support a grant application with a real web app and a deployed contract.

2. Problem

- New users on BOT Chain hold native BOT, but DEX pools trade WBOT, so a first swap needs a separate wrapping step in another tool.
- Routing across V3 fee tiers and multi-hop paths is invisible in basic interfaces, so users cannot tell if they got a fair price.
- Stale quotes, unchecked approvals, and failed transactions cost users gas and trust, especially inside mobile wallet browsers.
- Existing front ends rarely explain failures in plain language or protect a gas reserve for network fees.

3. Solution and positioning

BotFlow Exchange is a UX and safety layer on BDEX, not a new exchange. It creates no new liquidity and holds no user funds. Its job is to make the existing BDEX V3 pools on BOT Chain safe and simple to use: native BOT in one flow, transparent routing, exact-amount approvals with one click revoke, simulation before sending, and a re-quote right before signing. Everything runs in the browser with no backend, no tracking, and no custody of keys or funds.

4. System overview

A swap moves through five stages. (1) The web app quotes every candidate route on the BDEX V3 Quoter and picks the best one. (2) Price impact is measured against a trade 1/1000th the size on the same route. (3) The user confirms; only the steps needed are executed: wrap BOT to WBOT if paying with BOT, approve the router if allowance is low, re-quote and stop if the price fell below the minimum, simulate the swap, then send it with the slippage based minimum and the chosen deadline. (4) If the user receives BOT, the WBOT payout is unwrapped, measured from the balance change rather than the estimate. (5) Every transaction is signed in the user's own wallet and recorded in local recent activity with explorer links.

4.1 Routing policy

- Direct pools are quoted on every fee tier (100, 500, 3000, 10000 bps).
- Two hop routes through WBOT and USDT are quoted across 500, 3000 and 10000 bps legs.
- A direct pool is preferred unless a multi hop route is more than 0.1% better, since it uses less gas.
- BOT to WBOT and back is a direct 1:1 wrap with no pool fee.
- Quotes refresh every 15 seconds while the page is open and idle.

5. Web application

5.1 Protecting the user

- Fresh price check before signing: the route is quoted again and the swap stops before the wallet opens if the price fell below the user's minimum.

- Simulation before sending: every swap is statically simulated first, so a trade that would fail never costs gas.
- Price impact warnings: over 3% is highlighted, over 15% the button turns red and asks for extra confirmation.
- Readable errors: slippage, deadline, failed transfer, low gas, busy wallet and network faults are explained in plain language with a next step.
- Gas reserve: Max and the swap button leave 0.01 BOT in the wallet for network fees.
- Approvals manager: every token the router can spend is listed with one click revoke.

5.2 Settings and convenience

- Slippage presets 0.1%, 0.5%, 1% plus custom up to 50%, with warnings for extreme values.
- Transaction deadline from 1 to 180 minutes (default 20).
- Approval mode: exact amount per swap (default, safer) or unlimited.
- Network fee estimate in BOT before signing, shareable swap links, add token to wallet, per wallet recent activity with pending tracking across sessions.
- Works with MetaMask, OKX, Bitget, TokenPocket and other EVM wallets, including in app browsers. Mobile first with light and dark mode.

6. Smart contract: BotFlowRouter

BotFlowRouter (src/BotFlowRouter.sol, Solidity 0.8.24) is the on chain entry point for BotFlow swaps. It sits in front of the BDEX V3 SwapRouter and adds what a plain pool router leaves to the front end: native BOT in one transaction, single and multi hop routes through one function using the standard V3 packed path (token, fee, token, fee, token) with on chain layout validation, minimum output enforced on what the user actually receives (measured by balance change, checked after any protocol fee), support for fee on transfer input tokens by swapping only the amount that arrived, and zero funds held between transactions with router approvals reset in the same call. Stray BOT sent to the contract is rejected; only WBOT may send BOT during unwrapping.

Function	What it does
swapExactTokensForTokens(path, amountIn, minOut, recipient, deadline)	ERC-20 to ERC-20
swapExactBOTForTokens(path, minOut, recipient, deadline)	Native BOT to ERC-20, path starts with WBOT
swapExactTokensForBOT(path, amountIn, minOut, recipient, deadline)	ERC-20 to native BOT, path ends with WBOT
setFee / pause / unpause / rescue	Owner only

6.1 Fees

The protocol fee is taken from the swap output and starts at 0%. It is hard capped at 0.30% (30 bps) in the contract itself, so users can never be charged more, whatever the owner sets.

6.2 Safety controls

- OpenZeppelin ReentrancyGuard on all swap paths and Pausable for emergencies.
- Two step ownership transfer (Ownable2Step).
- Custom errors for clear revert reasons, deadline checks on every swap.

7. Testing and audit status

24 unit and fuzz tests in test/BotFlowRouter.t.sol cover every swap path, slippage and deadline checks, malformed paths, fee on transfer tokens, a reentrancy attack, fee maths and caps, pausing, ownership and rescue. Every test also asserts the contract is left holding no tokens, no BOT and no open approvals. Line coverage is about 97%. Fork tests in test/fork/BotChainFork.t.sol run a real tBOT to USDT round trip against live BDEX pools on a fork of BOT Chain Testnet. Status: the contract has NOT had an independent security audit. It is deployed on testnet only. An audit is required before any mainnet use, and mainnet ownership should be a multisig.

8. Network and deployments

Item	Value
Chain	BOT Chain Testnet (ID 968, 0x3c8)
RPC	https://rpc.bohr.life
Explorer	https://scan.bohr.life
Native coin	tBOT (18 decimals)
Faucet	https://faucet.botchain.ai/basic

Contract	Address
BotFlowRouter	0xEA00197fe5Ec5E5BcF2F6Bdc180d187c46c4aCD9
BDEX V3 SwapRouter	0x07032d47A1b9f8460cBeE9dC17c1d3E438693929
BDEX V3 QuoterV2	0x034A705b36067cff99ABf5C662Be881cBd8d0176
WBOT	0xD5452816194a3784dBa983426cCe7c122F4abd30
USDT (test)	0x75edC9335175Fc0552D51D48439F229c10420fe3

BotFlowRouter was deployed on 2026-10-06 by 0x65aC7753E03C216096e867e03898a6720B5449F9 (also the current owner), deploy transaction 0x5e9dbe0a6afd7ce09db97b505bf8f4dd805d05352e9e2b8e6cc3b2132d6c095d. Verify all addresses on the explorer before use.

9. Roadmap

- Switch the web app to route swaps through BotFlowRouter (contract deployed, integration next).
- Independent security review of BotFlowRouter before mainnet.
- Shared token list with verified logos and USD values next to amounts.
- Add liquidity and position views for BDEX V3 pools.
- Split routes across several pools for large trades.
- Translations, starting with Swahili.

10. Repository

Source: <https://github.com/Carter254g/BotFlow-Exchange> (MIT license). Layout: src/BotFlowRouter.sol (router contract), src/interfaces/ (BDEX router and WBOT interfaces), test/ (unit, fuzz and fork tests), script/Deploy.s.sol (deployment script), app/ (swap app), index.html + assets/ (landing page).

11. Disclaimer

BotFlow Exchange is an independent interface and is not affiliated with BOT Chain or BDEX. Test tokens have no real value. Nothing in this document is financial advice. Smart contracts carry risk; the router is unaudited and deployed on testnet only. Users must verify contract addresses on the explorer and never

share private keys or seed phrases with any app or person.